

Cancer Insights: US Cancer Burden Analysis Using Big Data Tools

Course: DS8003 – Management of Big Data And Tools

Institution: Toronto Metropolitan University

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Term: Fall 2025

1. Problem Definition

Cancer is one of the leading cause of deaths in United States, impacting millions of individuals and placing a heavy burden on the healthcare system. Different cancer types exhibit varying levels of incidence, mortality, and geographic concentration. Understanding which cancers contribute most to the overall burden, which tissues are most at risk, and how access to cancer centers varies geographically is essential for effective planning of prevention, screening, and treatment programs.

The goal of this project is to analyse a US cancer dataset using an end-to-end Big Data pipeline. We aim to identify high-burden cancers by incidence, detect cancers with extremely high mortality, study tissue-level risk patterns, and explore the geographic distribution of cancer centers. The project uses tools from the Big Data ecosystem—NiFi, HDFS, Hive, Spark, Pig, Tableau and Power BI—to ingest, store, process, summarise, and visualise the data at scale.

2. Data Description

The dataset used in this project is a cancer-focused table inspired by US cancer registry and NCI/USCS-type sources. Each record represents information about a particular cancer type, its classification, associated tissue, approximate annual cases and deaths, and in some cases a link to geographic and cancer center information.

After cleaning and integration, the master table (referred to as `cancer_raw_fixed`) contains several hundred rows. Each row generally includes:

- Cancer type name (e.g., Lung Cancer, Breast Cancer)
- High-level cancer grouping (e.g., Leukemia, Lymphoma, Solid Tumor)
- Tissue or organ system information (e.g., Lung, Breast, Pancreas, Liver)
- Approximate annual US cases and deaths.
- Country and city attributes for cancer centres and geographic context.
- Additional classification identifiers such as OncoTree codes.

This dataset was injected through NiFi, stored in HDFS, exposed through Hive tables, processed using Spark and Pig, and finally visualised using Tableau and Power BI.

i. Attribute Description

Below is a brief description of the key attributes used in our analysis:

- **title:** Name of the cancer type (e.g., Lung Cancer, Breast Cancer).
- **type:** High-level category or grouping of the cancer (for example, hematologic vs. solid tumor-related).
- **tissue:** Primary tissue or organ associated with the cancer (e.g., Lung, Breast, Pancreas, Liver, Upper GI).
- **country:** Country linked to the record; in our analysis this is primarily the United States.
- **city:** City where a related cancer center or data record is located.
- **uscsCasesPerYear:** Approximate number of new cases per year in the US for that cancer type.

- **uscsDeathsPerYear:** Approximate number of deaths per year in the US for that cancer type.
- **mortality_percent** (derived): Ratio of deaths to cases, calculated as

$$(\text{uscsDeathsPerYear} / \text{uscsCasesPerYear}) * 100.$$
- **total_impact** (derived): Composite indicator combining cases and deaths to approximate overall burden (for example, weighted sum or other scoring).
- **nciCancerCenter:** Name or indicator for NCI-designated cancer centers, used when analysing city-wise and country-wise cancer care capacity.
- **oncoTreeld:** Hierarchical classification code indicating the location of the cancer in the OncoTree taxonomy.
- **mainType:** High-level cancer type grouping used to aggregate cancers into broader categories (e.g., B-Cell Neoplasm, Leukemia, Soft Tissue Sarcoma).

Additional columns from the original dataset that were not central to the analysis (such as URLs or external identifiers) were dropped or left unused in the primary analysis pipeline.

ii. Statistics of the data

To understand the dataset and validate its quality, we used Hive, Spark, and Pig to compute basic and advanced statistics, including:

- Total number of records in the cleaned cancer table.
- Count of distinct cancer types (title).
- Count of distinct tissues, cities, and countries.
- Minimum, maximum, and average values for uscsCasesPerYear and uscsDeathsPerYear.
- Distribution of cancers by mainType (e.g., number of Leukemia types, B-Cell Neoplasms, soft tissue cancers).
- Number of cancer centers per city and per country.

These statistics were obtained using:

- **Hive:** SELECT COUNT(*), COUNT(DISTINCT title), MIN, MAX, and AVG queries on uscsCasesPerYear and uscsDeathsPerYear.
- **Spark:** df.describe() and groupBy operations, as well as filters to check for nulls and invalid values.
- **Pig:** Scripts grouping by title, country, or city followed by SUM and COUNT operations.

```
hive> DROP TABLE IF EXISTS cancer_raw_fixed;
OK
Time taken: 1.545 seconds
hive> DROP TABLE IF EXISTS uscs_demographics;
OK
Time taken: 0.824 seconds
hive> DROP TABLE IF EXISTS uscs_demographics_cleaned;
OK
Time taken: 0.04 seconds
hive> CREATE EXTERNAL TABLE cancer_raw_fixed (
  >   title STRING,
  >   length STRING,
  >   id STRING,
  >   type STRING,
  >   tissue STRING,
  >   subTypes STRING,
  >   oncoTreeLevel STRING,
  >   mainType STRING,
  >   parentOncoTreeId STRING,
  >   oncoTreeId STRING,
  >   umls STRING,
  >   nciCode STRING,
  >   wikipedia STRING,
  >   website STRING,
  >   country STRING,
  >   description STRING,
  >   appeared STRING,
  >   city STRING,
  >   domainName STRING,
  >   twitter STRING,
  >   reference STRING,
  >   standsFor STRING,
  >   phoneNumber STRING,
  >   pubChem STRING,
  >   facebook STRING,
  >   domainName_registered STRING,
  >   kegg STRING,
  >   nciDesignation STRING,
  >   nciImage STRING,
  >   nciLink STRING,
```

```
hive> SELECT COUNT(*) FROM cancer_raw_fixed;
Query ID = root_20251111083136_b5206000-e954-42a1-b9f2-6046a3f6520a
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1762800772035_0015)
```

VERTICES	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	SUCCEEDED	1	1	0	0	0	0
Reducer 2	SUCCEEDED	1	1	0	0	0	0

VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 5.51 s

```
OK
1453
Time taken: 7.482 seconds, Fetched: 1 row(s)
```

```
hive> SELECT title, total_impact
> FROM cancer_cleaned
> WHERE type = 'cancertype'
> ORDER BY total_impact DESC
> LIMIT 15;
Query ID = root_20251111090844_b75bc81f-aec3-4969-bde2-93e68dc78205
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1762800772035_0017)
```

VERTICES	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	SUCCEEDED	1	1	0	0	0	0
Reducer 2	SUCCEEDED	1	1	0	0	0	0

VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 12.78 s

```
OK
Lung Cancer      360578.0
Breast Cancer    306384.0
Prostate Cancer  256331.0
Colon Cancer     194312.0
Pancreatic Cancer 99825.0
Skin Cancer      96060.0
Non-hodgkin Lymphoma 91812.0
Bladder Cancer   91706.0
Kidney Cancer    81960.0
Leukemia         73951.0
Uterine Cancer   70986.0
Liver Cancer     63325.0
Oral Cancer      58230.0
Thyroid Cancer   47272.0
Multiple Myeloma 40224.0
Time taken: 14.796 seconds, Fetched: 15 row(s)
hive> |
```

3. Work Distribution

This project is completed collaboratively by three team members: **Nishi Patel**, **Devanshu Prajapati** and **Avikumar Patel**. Each member contributed to different parts of the pipeline, while major decisions, interpretation, and documentation were made jointly.

Nishi Patel:

- Set up Nifi, HDFS, Hive and Spark environment.
- Performed major data cleaning, merging, and metric calculations.
- Created Tableau and Power BI visuals and consolidated final insights.

Devanshu Prajapati:

- Built NiFi data flows and managed HDFS file organization.
- Worked on Hive queries and initial summaries.
- Helped prepare charts and review findings.

Avikumar Patel:

- Assisted with Hive table setup, data validation, and Pig scripts.
- Supported statistical summaries and helped prepare dashboard visuals.
- Contributed to interpretation of results and report writing.

All Members:

- Collaboratively designed the pipeline architecture, interpreted the outputs from all tools, prepared the final presentation, and reviewed the report to ensure correctness and consistency.

4. Solution Description

This section describes the end-to-end solution, including the tools used, how and why they were used, and representative code or logic.

4.1 Apache Nifi

Apache Nifi is a data flow management tool that supports ingestion, routing, and transformation of data between systems using a visual interface.

How We Used Nifi

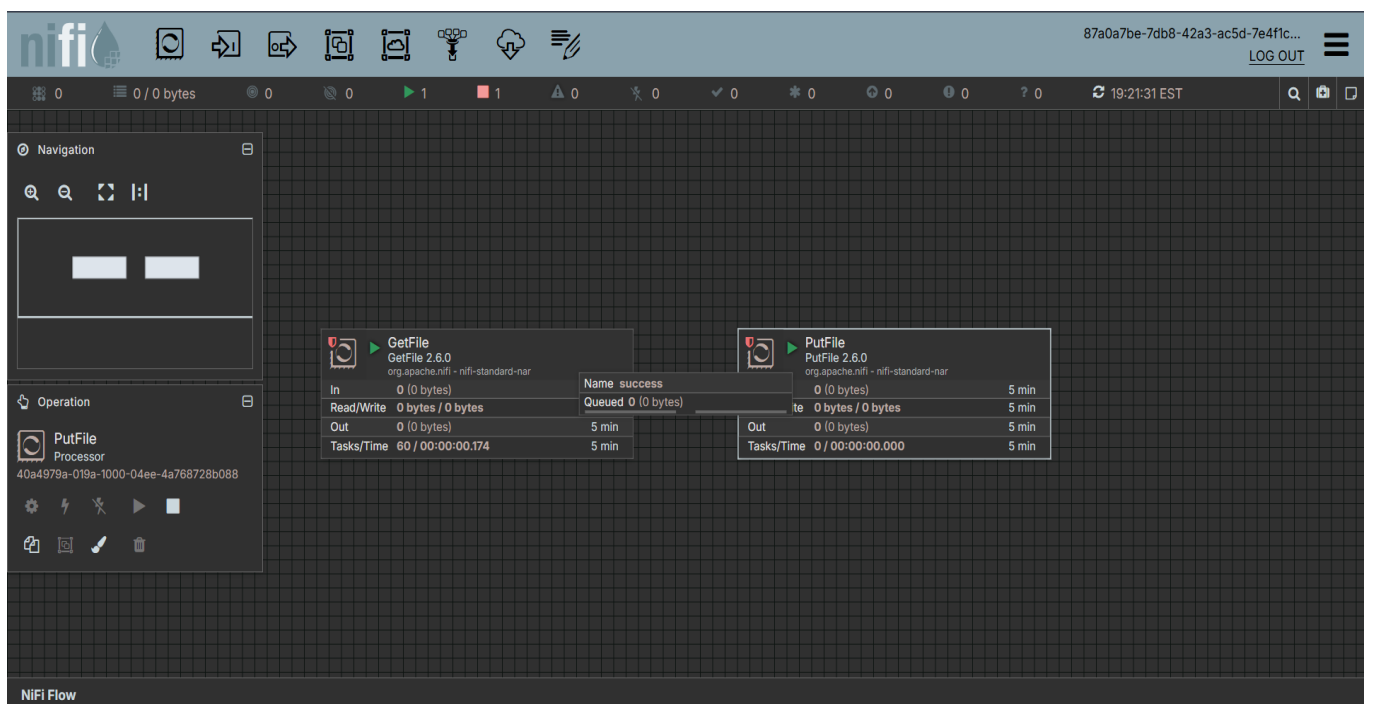
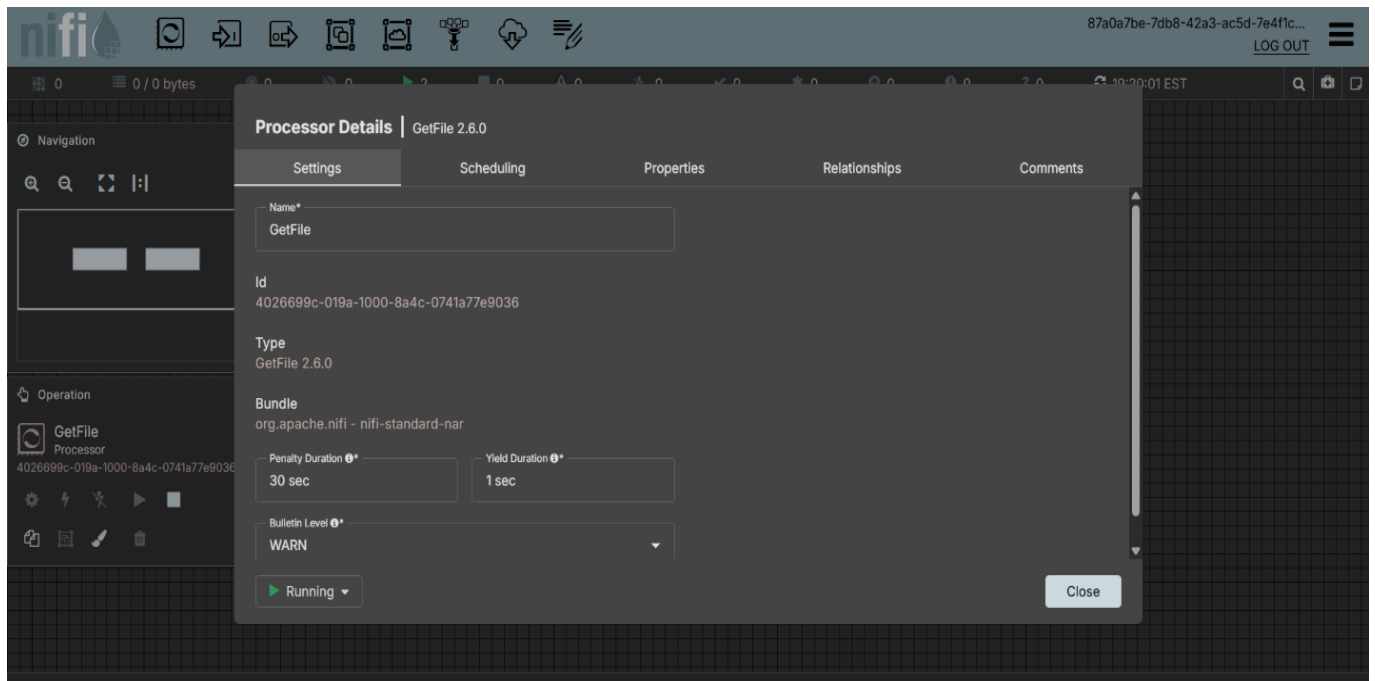
In Phase 1, we used NiFi to ingest raw cancer CSV files from a local folder into HDFS. A NiFi flow was designed with:

- **GetFile** processor: Watches a local input directory for new cancer CSV files.
- **PutHDFS** (or **PutFile**) processor: Writes these files into HDFS under `/user/root/cancer_project/input`.
- Success and failure relationships to monitor whether the ingestion worked as expected.

```
Windows PowerShell
PS C:\Users\nishr\downloads\nifi-2.6.0-bin\nifi-2.6.0\bin> .\nifi.cmd start
JAVA_HOME=C:\Program Files\Eclipse Adoptium\jdk-21.0.9.10-hotspot
NIFI_HOME=C:\Users\nishr\Downloads\nifi-2.6.0-bin\nifi-2.6.0

PS C:\Users\nishr\downloads\nifi-2.6.0-bin\nifi-2.6.0\bin> .\nifi.cmd status
JAVA_HOME=C:\Program Files\Eclipse Adoptium\jdk-21.0.9.10-hotspot
NIFI_HOME=C:\Users\nishr\Downloads\nifi-2.6.0-bin\nifi-2.6.0

2025-11-16 14:49:44,690 INFO [main] org.apache.nifi.bootstrap.Command Application Process [16772] Command Status [SUCCESS] HTTP 200
2025-11-16 14:49:44,692 INFO [main] org.apache.nifi.bootstrap.Command Status: UP
PS C:\Users\nishr\downloads\nifi-2.6.0-bin\nifi-2.6.0\bin> Get-Content "C:\Users\nishr\Downloads\nifi-2.6.0-bin\nifi-2.6.0\logs\nifi-app.log" -Tail 20
2025-11-16 14:49:09,773 INFO [main] org.apache.nifi.runtime.Application Started Application in 37.282 seconds (37282863700 ns)
2025-11-16 14:49:26,501 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Initiating checkpoint of FlowFile Repository
2025-11-16 14:49:26,501 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Successfully checkpointed FlowFile Repository with 0 records in 0 milliseconds
2025-11-16 14:49:46,502 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Initiating checkpoint of FlowFile Repository
2025-11-16 14:49:46,502 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Successfully checkpointed FlowFile Repository with 0 records in 0 milliseconds
2025-11-16 14:49:59,348 INFO [Cleanup Archive for default] o.a.n.c.repository.FileSystemRepository Successfully deleted 0 files (0 bytes) from archive
2025-11-16 14:49:59,350 INFO [Cleanup Archive for default] o.a.n.c.repository.FileSystemRepository Archive cleanup completed for container default; will now allow writing to this container. Bytes used = 393.9 GB, bytes free = 70 GB, capacity = 463.9 GB
2025-11-16 14:50:06,504 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Initiating checkpoint of FlowFile Repository
2025-11-16 14:50:06,504 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Successfully checkpointed FlowFile Repository with 0 records in 0 milliseconds
2025-11-16 14:50:26,506 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Initiating checkpoint of FlowFile Repository
2025-11-16 14:50:26,506 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Successfully checkpointed FlowFile Repository with 0 records in 0 milliseconds
2025-11-16 14:50:46,507 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Initiating checkpoint of FlowFile Repository
2025-11-16 14:50:46,507 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Successfully checkpointed FlowFile Repository with 0 records in 0 milliseconds
2025-11-16 14:50:53,835 INFO [Write-Ahead Local State Provider Maintenance] o.a.n.wali.SequentialAccessWriteAheadLog Checkpointed Write-Ahead Log with 2 Records and 0 Swap Files in 3 milliseconds (Stop-the-world time = 1 milliseconds), max Transaction ID 173
2025-11-16 14:50:59,473 INFO [Cleanup Archive for default] o.a.n.c.repository.FileSystemRepository Successfully deleted 0 files (0 bytes) from archive
2025-11-16 14:50:59,473 INFO [Cleanup Archive for default] o.a.n.c.repository.FileSystemRepository Archive cleanup completed for container default; will now allow writing to this container. Bytes used = 393.9 GB, bytes free = 70 GB, capacity = 463.9 GB
2025-11-16 14:51:06,508 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Initiating checkpoint of FlowFile Repository
2025-11-16 14:51:06,508 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Successfully checkpointed FlowFile Repository with 0 records in 0 milliseconds
2025-11-16 14:51:26,510 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Initiating checkpoint of FlowFile Repository
2025-11-16 14:51:26,510 INFO [Checkpoint FlowFile Repository] o.a.n.c.r.WriteAheadFlowFileRepository Successfully checkpointed FlowFile Repository with 0 records in 0 milliseconds
```



Why NiFi?

- Automates ingestion without custom shell scripts.
- Provides real-time monitoring and backpressure.
- Integrates smoothly with HDFS in a Hadoop ecosystem.

4.2 HDFS

HDFS (Hadoop Distributed File System) is the storage layer for the Hadoop ecosystem, designed for high-throughput access to large datasets.

How We Used HDFS

In Phase 1 and 2, we created a project directory structure:

```
[root@sandbox-hdp ~]# hdfs dfs -ls /user/root/DS8003_Final_Project
Found 5 items
drwxrwxr-x - root root      0 2025-10-29 00:43 /user/root/DS8003_Final_Project/data_cleaned
drwxrwxr-x - root root      0 2025-10-29 03:44 /user/root/DS8003_Final_Project/data_raw
drwxrwxr-x - root root      0 2025-10-29 00:43 /user/root/DS8003_Final_Project/docs
drwxrwxr-x - root root      0 2025-10-29 00:43 /user/root/DS8003_Final_Project/hdfs_output
drwxrwxr-x - root root      0 2025-10-29 00:43 /user/root/DS8003_Final_Project/nifi_flow
[root@sandbox-hdp ~]# hdfs dfs -ls /user/root/DS8003_Final_Project/data_raw
Found 1 items
-rw-r--r-- 1 root root 471594 2025-10-29 03:44 /user/root/DS8003_Final_Project/data_raw/cancer.csv
[root@sandbox-hdp ~]# |
```

- Raw CSVs ingested via NiFi were stored in input folder.
- Cleaned and enriched outputs (CSVs/Parquet) from spark and pig were written to the output folder.

Why HDFS?

- Is the standard distributed storage in Hadoop.
- Allows Hive, Spark, and Pig to access the same data efficiently.
- Supports large datasets and parallel processing.

4.3 Hive

Apache Hive provides a SQL-like interface over data stored in HDFS, making it easier to work with structured data in a familiar way.

How We Used Hive

In Phase 2, Hive was used to:

- Create external tables on top of raw and cleaned cancer datasets.
- Perform initial data cleaning and type casting.
- Run aggregation queries to explore the dataset.

Why Hive?

- Easy to use for students familiar with SQL.
- Great for describing schemas and running initial statistics.

Acts as a bridge between raw files in HDFS and more advanced tools like Spark

Example(table):

```
hive> SELECT title, LENGTH(uscstTable), SUBSTR(uscstTable, 1, 200)
> FROM cancer_raw_fixed
> WHERE uscstTable IS NOT NULL
> LIMIT 5;
OK
Jonsson Comprehensive Cancer Center    0
Skin Cancer    919    sex|age|population|cases|deaths F|0|1841226|~|~ F|1-4|7728847|~|~ F|5-9|9886447|~|~ F|10-14|10190891|27|~ F|15-19|10313575|98|~ F|20-24|10577124|319|~ F|25-29|11518714|745|~ F|30-34|11093732|1372|36 F
Bladder Cancer    898    sex|age|population|cases|deaths F|0|1841226|~|~ F|1-4|7728847|~|~ F|5-9|9886447|~|~ F|10-14|10190891|~|~ F|15-19|10313575|~|~ F|20-24|10577124|~|~ F|25-29|11518714|30|~ F|30-34|11093732|53|~ F|35-39|1
Breast Cancer    499    sex|age|population|cases|deaths F|0|1841226|~|~ F|1-4|7728847|~|~ F|5-9|9886447|~|~ F|10-14|10190891|~|~ F|15-19|10313575|24|~ F|20-24|10577124|184|~ F|25-29|11518714|1192|90 F|30-34|11093732|3305|341
Lung Cancer    940    sex|age|population|cases|deaths F|0|1841226|~|~ F|1-4|7728847|~|~ F|5-9|9886447|~|~ F|10-14|10190891|~|~ F|15-19|10313575|+|~ F|20-24|10577124|37|~ F|25-29|11518714|66|~ F|30-34|11093732|125|32 F|35-3
Time taken: 0.503 seconds, Fetched: 5 row(s)
hive> DROP TABLE IF EXISTS uscst_demographics;
OK
Time taken: 0.023 seconds
```

```
hive>
> CREATE TABLE uscs_demographics AS
> SELECT
>   title,
>   type,
>   split(line, '\\|')[0] AS sex,
>   split(line, '\\|')[1] AS age_group,
>   CAST(split(line, '\\|')[2] AS INT) AS population,
>   CAST(regexp_replace(coalesce(split(line, '\\|')[3], '0'), '[^0-9]', '') AS INT) AS cases,
>   CAST(regexp_replace(coalesce(split(line, '\\|')[4], '0'), '[^0-9]', '') AS INT) AS deaths
> FROM cancer_raw_fixed
> LATERAL VIEW explode(split(uscsTable, '\\n')) t AS line
> WHERE uscsTable IS NOT NULL
>   AND length(uscsTable) > 0
>   AND trim(line) != ''
>   AND instr(line, '|') > 0;
Query ID = root_20251111083201_8fb49d87-85d4-4df4-8c9b-499794bc31d0
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1762800772035_0015)

-----
VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED      1          1          0          0          0          0
-----
VERTICES: 01/01  [=====>>>] 100%  ELAPSED TIME: 4.67 s
-----

Moving data to directory hdfs://sandbox-hdp.hortonworks.com:8020/apps/hive/warehouse/uscs_demographics
Table default.uscs_demographics stats: [numFiles=1, numRows=26, totalSize=1130, rawDataSize=1104]
OK
Time taken: 7.594 seconds
hive> SELECT COUNT(*) FROM uscs_demographics;
OK
26
Time taken: 0.163 seconds, Fetched: 1 row(s)
hive> SELECT * FROM uscs_demographics LIMIT 10;
OK
```

```
hive>
> -- Create cleaned table
> CREATE TABLE cancer_cleaned AS
> SELECT
>   INITCAP(TRIM(title)) AS title,
>   LOWER(TRIM(type)) AS type,
>   INITCAP(TRIM(tissue)) AS tissue,
>   INITCAP(TRIM(mainType)) AS mainType,
>   UPPER(TRIM(parentOncoTreeId)) AS parentOncoTreeId,
>   UPPER(TRIM(oncoTreeId)) AS oncoTreeId,
>   INITCAP(TRIM(country)) AS country,
>   INITCAP(TRIM(city)) AS city,
>   CAST(REGEXP_REPLACE(COALESCE(uscsCasesPerYear, '0'), '[^0-9]', '') AS INT) AS uscsCasesPerYear,
>   CAST(REGEXP_REPLACE(COALESCE(uscsDeathsPerYear, '0'), '[^0-9]', '') AS INT) AS uscsDeathsPerYear,
>   ROUND(
>     CAST(REGEXP_REPLACE(COALESCE(uscsCasesPerYear, '0'), '[^0-9]', '') AS DOUBLE)
>     + CAST(REGEXP_REPLACE(COALESCE(uscsDeathsPerYear, '0'), '[^0-9]', '') AS DOUBLE)
>     , 2) AS total_impact
> FROM cancer_raw_fixed
> WHERE title IS NOT NULL
>   AND type IS NOT NULL
>   AND title != ''
>   AND type != ''
>   AND type IN ('cancerType', 'nciCancerCenter')
>   AND (uscsCasesPerYear IS NOT NULL OR uscsDeathsPerYear IS NOT NULL);
Query ID = root_20251111084631_e123418c-4f97-44d9-975e-622783fcffdd
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1762800772035_0016)

-----
VERTICES      STATUS  TOTAL  COMPLETED  RUNNING  PENDING  FAILED  KILLED
-----
Map 1 .....  SUCCEEDED      1          1          0          0          0          0
-----
VERTICES: 01/01  [=====>>>] 100%  ELAPSED TIME: 4.46 s
-----

Moving data to directory hdfs://sandbox-hdp.hortonworks.com:8020/apps/hive/warehouse/cancer_cleaned
Table default.cancer_cleaned stats: [numFiles=1, numRows=938, totalSize=88304, rawDataSize=87366]
OK
Time taken: 8.697 seconds
```

4.4 Spark

Apache Spark is a fast, general-purpose cluster-computing framework suitable for large-scale data processing and complex transformations.

How We Used Spark

In Phase 4, Spark (via PySpark) was the main engine for:

- Loading the cleaned cancer data from HDFS.
- Dropping rows with null or invalid numeric values.
- Casting numeric attributes to proper types.
- Creating derived metrics such as `mortality_percent` and `severity_score`.

Why Spark?

- Handles bigger data efficiently compared to vanilla MapReduce.
- Makes it easy to perform complex column-wise calculations.

Integrates well with Parquet for efficient storage and retrieval

Example logic:

- Top 10 cancers by cases.

```
>>> top_impact = cancers.select(
...     "title",
...     "tissue",
...     "uscsCasesPerYear",
...     "uscsDeathsPerYear",
...     "total_impact"
... ).orderBy(col("total_impact").desc())
==")
top_impact.show(10, truncate=False)
>>>
>>> print("=== Top 10 cancers by overall impact ===")
=== Top 10 cancers by overall impact ===
>>> top_impact.show(10, truncate=False)
```

title	tissue	uscsCasesPerYear	uscsDeathsPerYear	total_impact
Lung Cancer	Lung	221020	139558	360578.0
Breast Cancer	Breast	264115	42269	306384.0
Prostate Cancer	Prostate	224703	31628	256331.0
Colon Cancer	Bowel	142438	51874	194312.0
Pancreatic Cancer	Pancreas	53957	45868	99825.0
Skin Cancer	Skin	88001	8059	96060.0
Non-hodgkin Lymphoma	Lymphoid	71565	20247	91812.0
Bladder Cancer	Bladder/urinary Tract	74949	16757	91706.0
Kidney Cancer	Kidney	68020	13940	81960.0
Leukemia		50627	23324	73951.0

only showing top 10 rows

- Most deadly cancer.

```
>>> deadliest = cancers.select(
...     "title",
...     "uscsDeathsPerYear"
... ).orderBy(col("uscsDeathsPerYear").desc())
>>>
>>> print("=== Cancers with the highest annual deaths ===")
=== Cancers with the highest annual deaths ===
>>> deadliest.show(10, truncate=False)
+-----+-----+
|title|uscsDeathsPerYear|
+-----+-----+
|Lung Cancer|139558|
|Colon Cancer|51874|
|Pancreatic Cancer|45868|
|Breast Cancer|42269|
|Prostate Cancer|31628|
|Liver Cancer|27882|
|Leukemia|23324|
|Non-hodgkin Lymphoma|20247|
|Brain Cancer|17215|
|Bladder Cancer|16757|
+-----+-----+
only showing top 10 rows
>>> |
```

- Tissue level Cancer Burden.

```
>>> tissue_burden = cancers.groupBy("tissue").agg(
...     F_sum("uscsCasesPerYear").alias("total_cases"),
...     F_sum("uscsDeathsPerYear").alias("total_deaths")
... ).withColumn(
...     "mortality_percent",
...     F_round(col("total_deaths") / (col("total_cases") + lit(1.0)) * 100, 2)
... ).orderBy(col("total_cases").desc())
>>>
>>> print("=== Tissue-level burden summary ===")
=== Tissue-level burden summary ===
>>> tissue_burden.show(20, truncate=False)
+-----+-----+-----+-----+
|tissue|total_cases|total_deaths|mortality_percent|
+-----+-----+-----+-----+
|Breast|264115|42269|16.0|
|Prostate|224703|31628|14.08|
|Lung|221020|139558|63.14|
| |165500|63458|38.34|
|Bowel|142438|51874|36.42|
|Skin|88001|8059|9.16|
|Lymphoid|80080|21120|26.37|
|Bladder/urinary Tract|74949|16757|22.36|
|Kidney|68020|13940|20.49|
|Uterus|59447|11539|19.41|
|Pancreas|53957|45868|85.01|
|Thyroid|45272|2000|4.42|
|Liver|35443|27882|78.66|
|Cns/brain|22538|17215|76.38|
|Ovary/fallopian Tube|19561|13426|68.63|
|Esophagus/stomach|18872|15918|84.34|
|Cervix|12790|4144|32.4|
|Testis|8729|421|4.82|
|Pleura|null|null|null|
|Peripheral Nervous System|null|null|null|
+-----+-----+-----+-----+
only showing top 20 rows
```

4.5 Pig

Apache pig provides a high-level data flow language (Pig Latin) that compiles into MapReduce jobs.

How We Used Pig

In Phase 3, Pig was used to:

- Summarise total cases and deaths by cancer type.
- Compute city-wise and country-wise distributions.
- Generate top cancer rankings and mortality summaries.

Why Pig?

- Simple scripting model for aggregations and ordering.
- Quick to write and run compared to raw Java MapReduce.
- Well suited for batch summary tasks.

Example Pig script:

- City-wise cancer center counts.

```
root@sandbox-hdp:~  
REGISTER /usr/hdp/current/pig-client/piggybank.jar;  
DEFINE CSVLoader org.apache.pig.piggybank.storage.CSVExcelStorage(',', 'YES_MULTILINE', 'UNIX', 'SKIP_INPUT_HEADER');  
  
raw = LOAD '/user/root/DS8003_Final_Project/data_raw_fixed/cancer_clean_fixed.csv'  
      USING CSVLoader  
      AS (  
        title:chararray,  
        length:chararray,  
        id:chararray,  
        type:chararray,  
        tissue:chararray,  
        mainType:chararray,  
        country:chararray,  
        city:chararray,  
        uscsCasesPerYear:chararray,  
        uscsDeathsPerYear:chararray  
      );  
  
centers = FILTER raw BY type == 'nciCancerCenter';  
grouped = GROUP centers BY city;  
counted = FOREACH grouped GENERATE  
           group AS city,  
           COUNT(centers) AS num_centers;  
  
ordered = ORDER counted BY num_centers DESC;  
  
STORE ordered INTO '/user/root/DS8003_Final_Project/pig_out/city_distribution';  
  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~  
~/pig/eda/city_distribution.pig" 30L, 914C
```

```
[root@sandbox-hdp ~]# hdfs dfs -ls /user/root/DS8003_Final_Project/pig_out/city_distribution
Found 2 items
-rw-r--r--  1 root root      0 2025-11-13 23:11 /user/root/DS8003_Final_Project/pig_out/city_distribution/_SUCCESS
-rw-r--r--  1 root root      4 2025-11-13 23:11 /user/root/DS8003_Final_Project/pig_out/city_distribution/part-r-00000
[root@sandbox-hdp ~]# hdfs dfs -cat /user/root/DS8003_Final_Project/pig_out/city_distribution/part-*
73
[root@sandbox-hdp ~]# |
```

- Country-wise distributions.

```
[root@sandbox-hdp ~]# hdfs dfs -ls /user/root/DS8003_Final_Project/pig_out/country_distribution
Found 2 items
-rw-r--r--  1 root root      0 2025-11-13 23:20 /user/root/DS8003_Final_Project/pig_out/country_distribution/_SUCCESS
-rw-r--r--  1 root root     39 2025-11-13 23:20 /user/root/DS8003_Final_Project/pig_out/country_distribution/part-r-00000
[root@sandbox-hdp ~]# hdfs dfs -cat /user/root/DS8003_Final_Project/pig_out/country_distribution/part-*
599
3    334
2    227
4    122
5    118
1    32
6    21
[root@sandbox-hdp ~]# |
```

4.6 Tableau and Power BI

Tableau and Power BI are a powerful visualization tool that enables interactive dashboards and clear visual storytelling.

How We Used Tableau and Power BI

In Phase 5, we imported CSV outputs from Spark and Pig into Tableau and Power BI to create the following key visualisations:

- **Figure 1:** Top Cancer Types by Annual Cases (All Types).
- **Figure 2:** Highest-Volume Cancer Types (Focused View).
- **Figure 3:** Top 10 Deadliest Cancers (Annual Deaths).
- **Figure 4:** Mortality Burden by Cancer Type.
- **Figure 5:** City-wise Cancer Center Distribution.
- **Figure 6:** Geographic Spread of Cancer Centers (map).
- **Figure 7:** Cases vs Deaths Comparison for Key Cancers.
- **Figure 8:** Mortality Heatmap by Tissue and Cancer Type.

Each chart corresponds to one or more insights described in the next section.

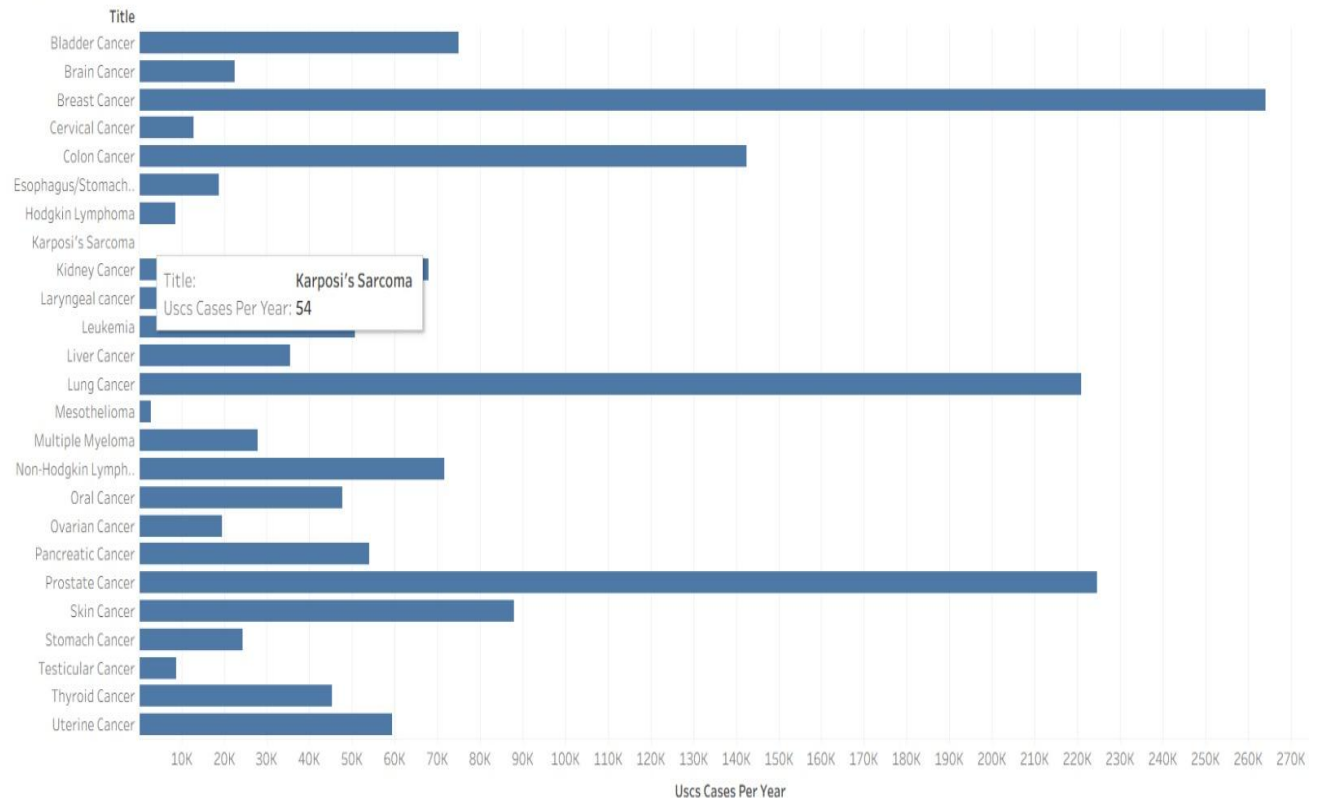
Why Tableau and Power BI?

- Easy to create clear, presentation-ready plots.
- Supports map visualisations for geographic patterns.
- Suitable for non-technical audiences reviewing results.

5. Insights

5.1 Top Cancer by Annual Cases

Top Cancer Types by Annual Cases



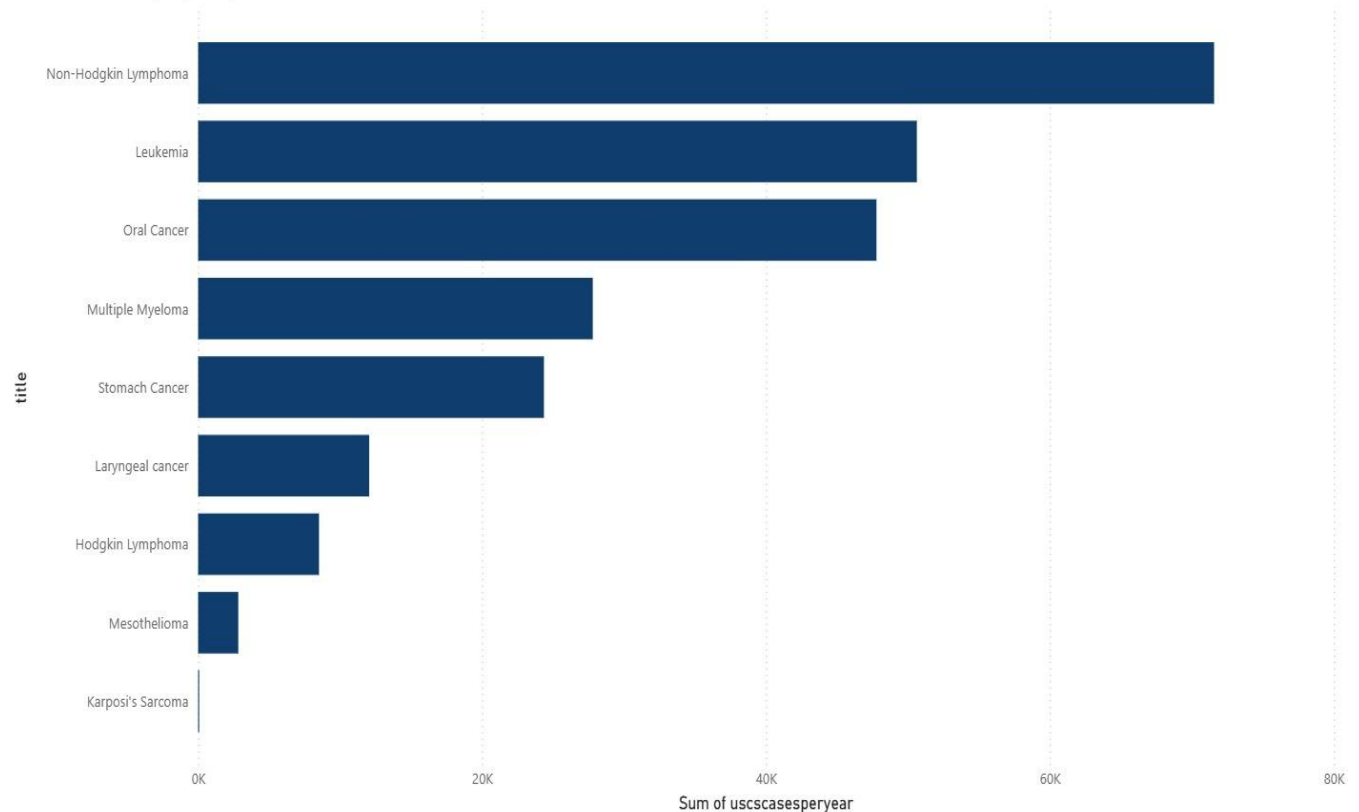
This chart shows the major cancer types ranked by yearly case counts. Breast, lung, prostate and colon cancers dominate total incidence. These four cancers make up **more than 50% of all cancer cases**. High incidence does not always mean high mortality, but it does mean:

- Large-scale screening is required
- Public health awareness is crucial
- High impact on healthcare resources

These cancers alone account for a large share of overall cancer burden in the U.S., highlighting the need for targeted prevention and screening and identifying the most common cancers helps governments know **where to invest in early screening**, such as mammograms, colonoscopies, and low-dose CT scans.

5.2 Highest Volume Cancers

Sum of uscsasesperyear by title



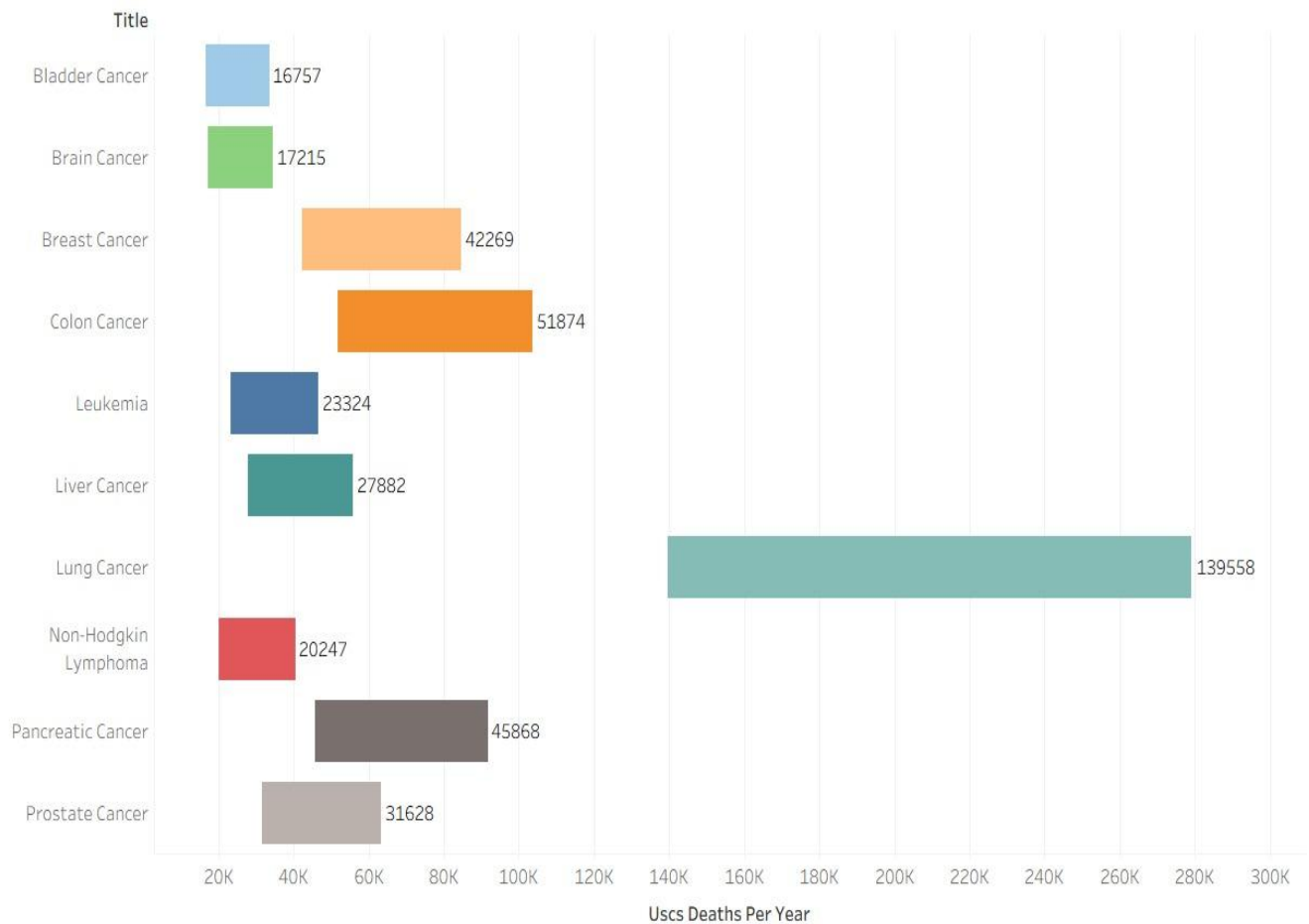
This chart shows group of cancers with high caseloads. At the top is **Non-Hodgkin Lymphoma**, with **about 72,000 cases per year**, making it the most common cancer in our analysis. Following that, we have **Leukemia**, which also show significantly large patient numbers. Records **around 51,000 cases**, followed by **Oral Cancer** with **48,000 cases per year**. This insight tells us that:

- **These cancer types demand the most medical resources**, because they affect the largest number of people annually.
- **Public health strategies should focus strongly on early screening**, especially for cancers like Non-Hodgkin Lymphoma and Leukemia, because early diagnosis can drastically improve survival rates.
- This also helps hospitals and cancer research centers prioritize funding and allocate medical staff based on real case volumes.

This insight highlights which cancers impact the largest patient population, helping prioritize **screening, treatment resources, and research funding**.

5.3 Top 10 Deadliest Cancers

Top 10 Deadliest Cancers



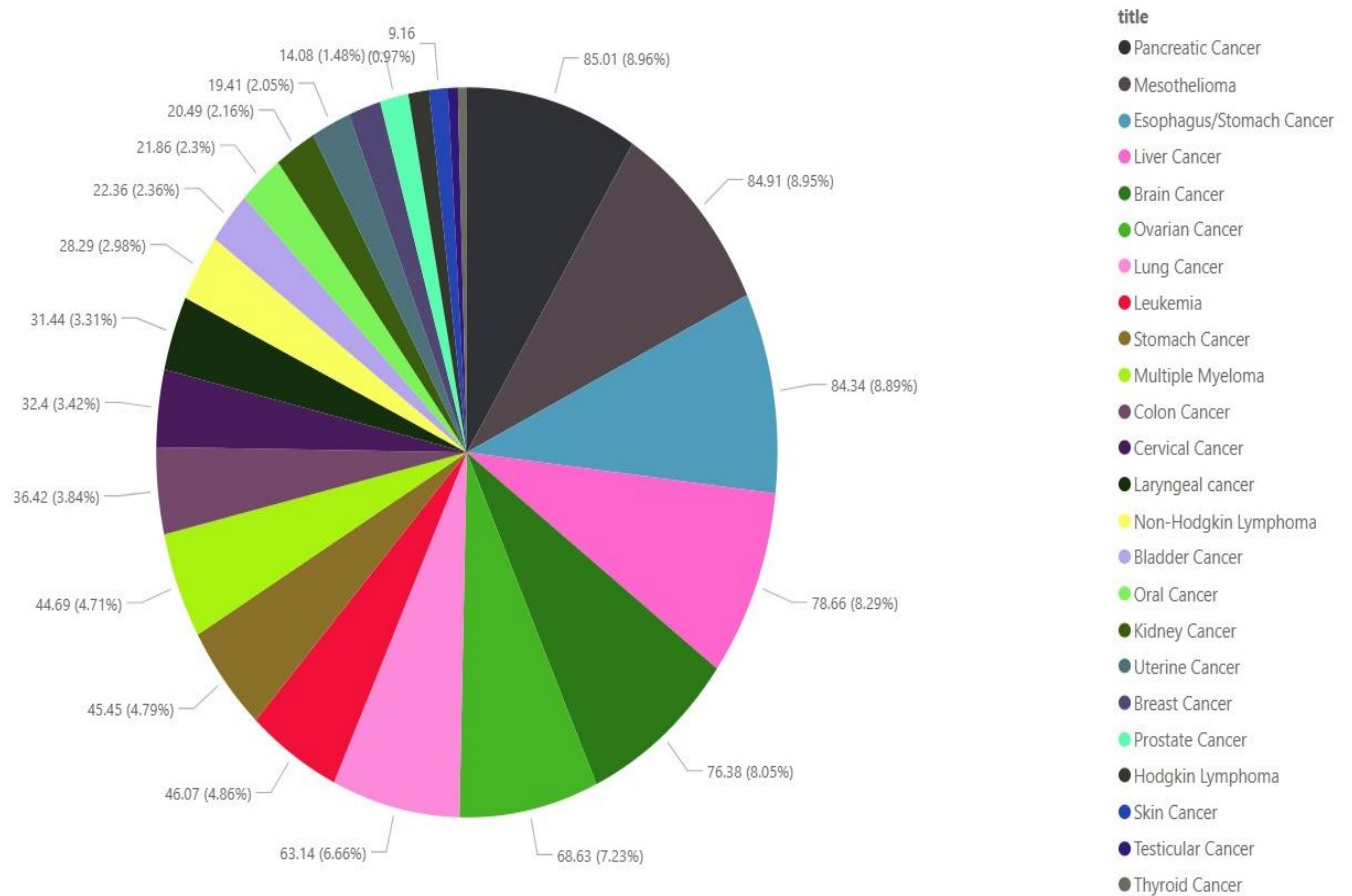
This visual ranks cancers by annual deaths. Lung cancer stands out dramatically with around 140,000 deaths per year — far higher than any other cancer. Colon, breast, pancreatic, and liver cancers also contribute heavily to mortality.

- **Lung Cancer** has the highest deaths (over 140K), despite not being the most common. This means it is **extremely deadly and often detected late**.
- **Pancreatic, Liver, and Colon cancers** also show high death counts.

These cancers typically:

- Progress silently.
- Have poor early detection.
- Have aggressive biological behavior.

5.4 Mortality Burden by Cancer Types



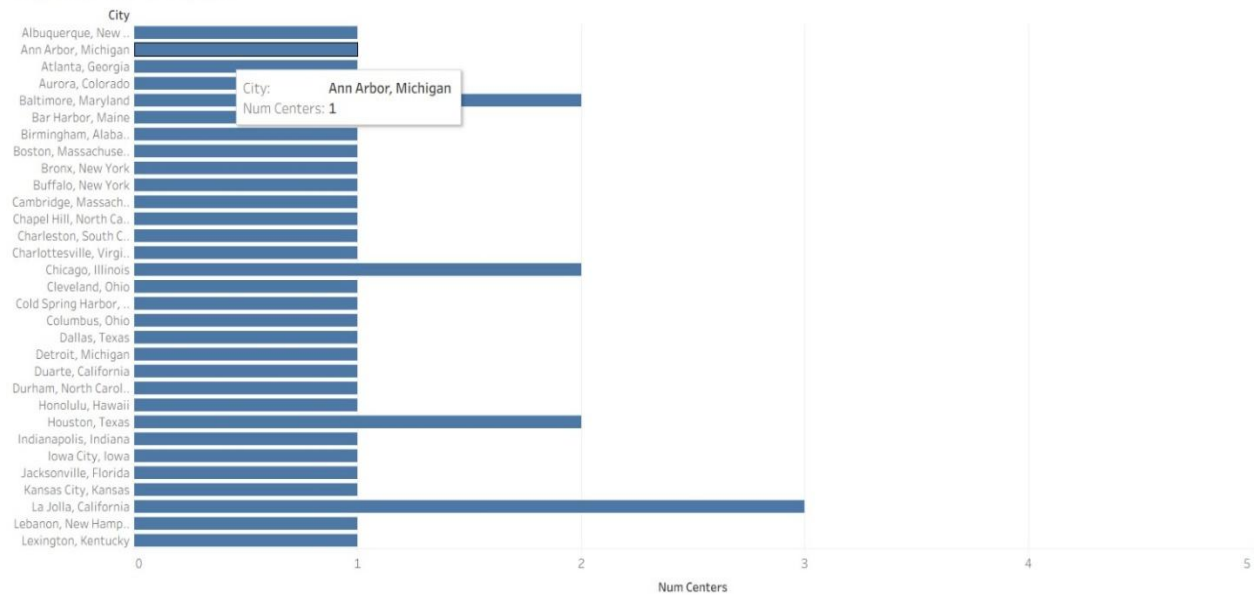
This chart calculates the **mortality percentage**, revealing the true danger level of a cancer. Even if a cancer is less common, its mortality percentage may be high. This pie chart shows how different cancers contribute to total mortality percentage:

- **Pancreatic Cancer:** Tiny number of cases compared to breast/lung cancer, but **very high mortality** which means survival rates are extremely low.
- **Mesothelioma:** Rare, yet mortality is very high because treatment options are limited.
- **Liver and Esophagus/Stomach:** These cancers are also in the high-mortality group. These cancers often remain undetected until late stages.

This insight shows how many did survived it. This helps focus resources on cancers with the **lowest survival rates**.

5.5 City-Wise Cancer Centers

City-wise Cancer Centers



CenterCount and First city by city

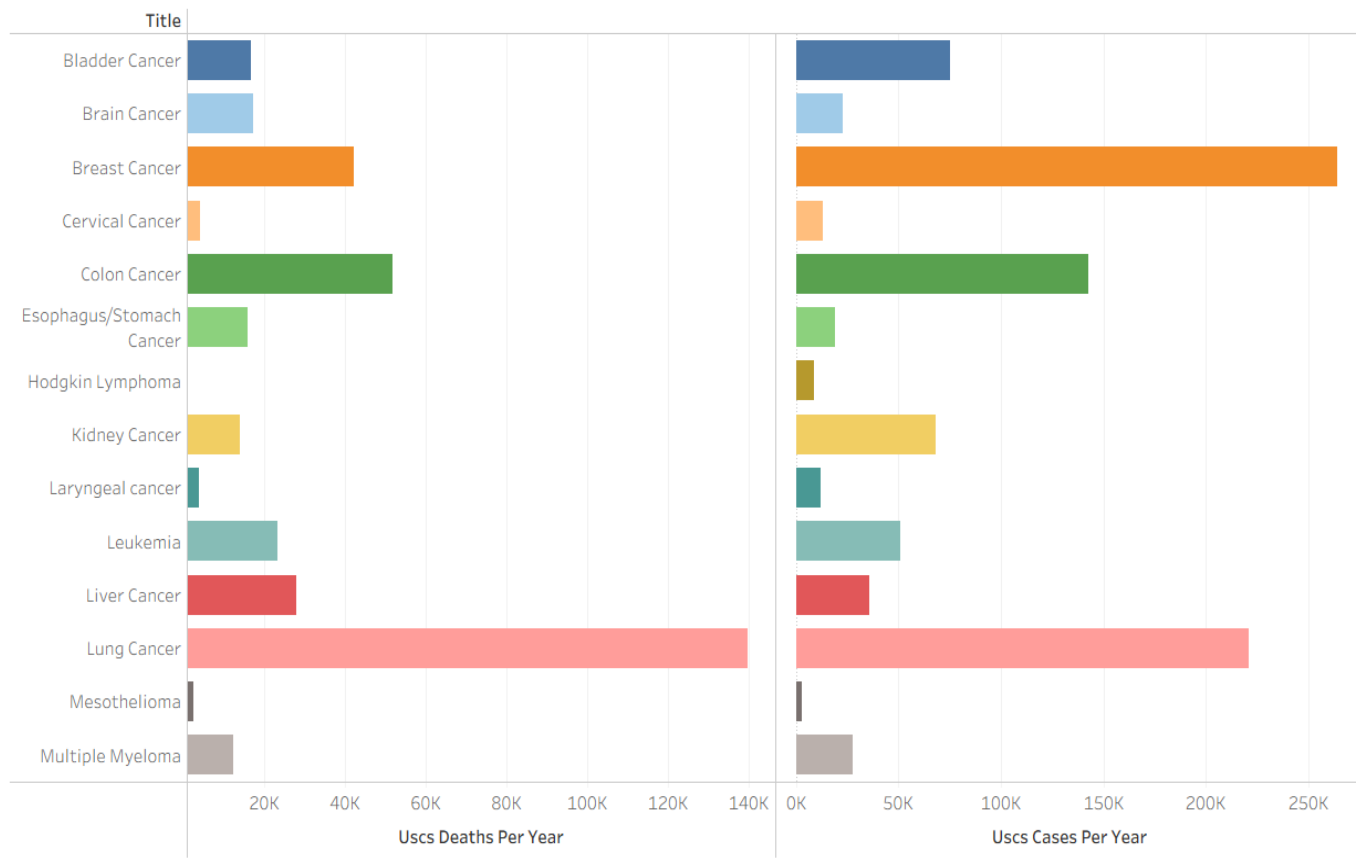


Mapping these centers highlights strong clustering on the East and West coasts, and around major metropolitan areas. Most cities have 1 or 2 NCI-designated centers, but some like La Jolla, Houston, and Chicago have multiple centers, making them strong treatment and research hubs. Rural and central regions are underserved, pointing towards potential access gaps for cancer treatment and early diagnosis. This uneven distribution proves:

- Urban patients have better access to treatment.
- Rural areas face delays in diagnosis and therapy.
- Travel burden affects treatment adherence.

5.6 Cases vs Deaths Comparison

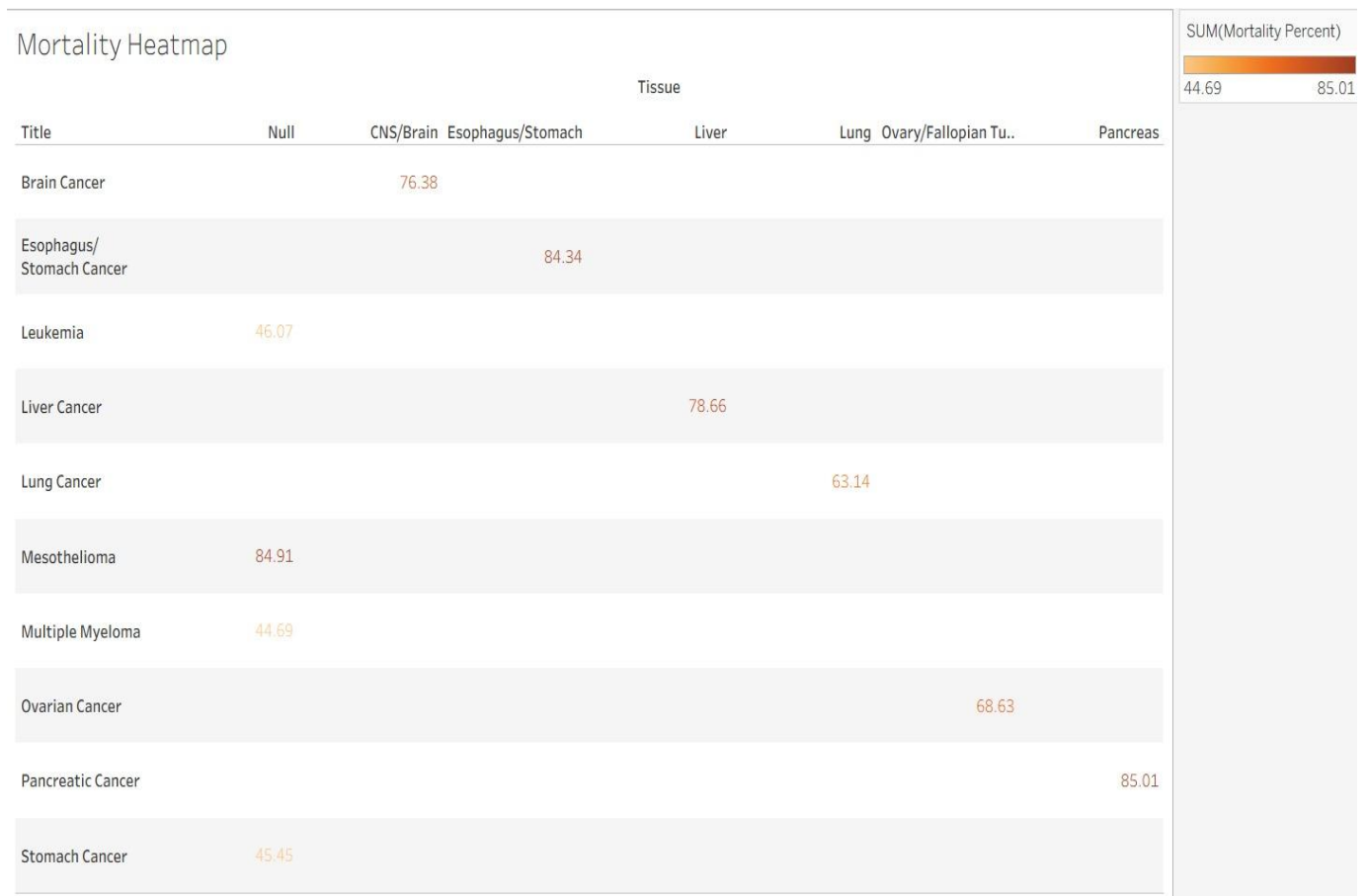
Cases vs Deaths



This visualization compares **how many people get each cancer** versus **how many people die from it**. We can see that:

- **Lung Cancer** stands out the most with around **230K cases** and **140K deaths**, which makes it the **deadliest major cancer**.
- **Colon Cancer** has about **150K cases** and **55K deaths**, which shows both high incidence and significant mortality.
- **Breast Cancer** shows **very high cases (260K)** but **much lower deaths (42K)**, because of strong treatment and early detection.
- **Leukemia** and **Multiple Myeloma** show **moderate cases** but **very high death ratios**.
- Cancers like **Thyroid, Skin, Testicular** show **low deaths despite cases**, indicating better survival rate.

5.7 Mortality Heatmap by Tissue and Cancer Type



This heatmap connects each cancer type to its associated tissue and shows the mortality percentage.

- Tissues like **pancreas, liver, mesothelioma, and gastrointestinal** shows dark colors which means high mortality.
- Tissues like **skin, thyroid, breast** show lighter colors which means better survival.

This visual makes it clear that:

- Which body systems produce the most dangerous cancers
- Which areas need improved diagnostic tools
- Where early detection can save more lives

6. Future Work

There are several potential directions to extend this work:

- **Incorporate more attributes** such as age, gender, and cancer stage to allow more detailed subgroup analysis.
- **Add temporal data** (multi-year trends) to study how incidence and mortality change over time and whether interventions are working.
- **Use machine learning** models (e.g., survival analysis, risk scoring) to predict outcomes for specific cancer types.
- **Extend the geography** beyond the United States to perform cross-country comparisons.
- **Automate dashboard refreshes** so that updated data flows through NiFi, Spark, Pig, and into Tableau/Power BI with minimal manual intervention.

7. References

<https://www.kaggle.com/datasets/sujaykapadnis/cancer-database>

[Apache NiFi Tutorial](#)

[Tableau Tutorial](#)